# STUDENT CLASSIFICATION SYSTEM

## MACHINE LEARNING MODEL PERFORMANCE REPORT

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## 1. Model Information

| Parameter | Value |
|---|---|
| Best Model | Gradient Boosting |
| Target Variable | TARGET_RISK |
| Dataset Records | 12632 |
| Number of Features | 16 |

## 2. Overall Performance

| Evaluation Metric | Score | Percentage |
|---|---|---|
| Test Accuracy | 0.9786 | 97.86% |
| Cross Validation Macro F1 | 0.9074 | 90.74% |
| Test Macro F1 | 0.8872 | 88.72% |

Accuracy measures the proportion of correct predictions. Precision measures the correctness of positive predictions. Recall measures how many actual positive cases were identified. F1-score combines precision and recall.

## 3. Class-wise Precision, Recall and F1 Score

| Class | Precision | Recall | F1 Score | Support |
|---|---|---|---|---|
| 0 | 0.8684 | 0.7174 | 0.7857 | 138.0 |
| 1 | 0.9838 | 0.9937 | 0.9888 | 2389.0 |
| macro avg | 0.9261 | 0.8556 | 0.8872 | 2527.0 |
| weighted avg | 0.9775 | 0.9786 | 0.9777 | 2527.0 |

## 4. Confusion Matrix

| Actual / Predicted | 0 | 1 |
|---|---|---|
| 0 | 99 | 39 |
| 1 | 15 | 2374 |

## 5. Evaluation Notes

The metrics in this report are taken from the saved model evaluation metadata. Test metrics represent the held-out test set used during model training. Cross-validation results represent the saved cross-validation score. Results should be interpreted with the dataset size, class distribution and evaluation methodology in mind.